C3.1 Read and write address channel signaling

The following sections define the additional signals on the read and write address channels.

C3.1.1 Shareability domain types

The ACE protocol uses a concept called shareability domains. A shareability domain is a set of masters that enables a master to determine which other masters to include when issuing coherency or barrier transactions. See *Domains on page C1-147*

Each address channel has its own shareability domain signal. Table C3-1 shows the signal for each address channel.

Table C3-1 Shareability domain signals

Transaction Channel	Signal	Source	Description
Read address channel	**ARDOMAIN[1:0]**	Master	Indicates the shareability domain of a read transaction.
Write address channel	**AWDOMAIN[1:0]**	Master	Indicates the shareability domain of a write transaction.

The ACE protocol specifies four levels of shareability domain:

Non-shareable The domain contains a single master.

Inner Shareable The Inner domain can include additional masters.

Outer Shareable The Outer domain contains all masters in the Inner domain and can include additional masters.

System The System domain includes all masters in the system.

Note

Although multiple Non-shareable, Inner Shareable and Outer Shareable domains can exist in a system, there must be a single consistent definition of the masters that are contained in each domain. Figure C1-3 on page C1-147 shows an example set of shareability domains.

In this specification, **AxDOMAIN** indicates **ARDOMAIN** or **AWDOMAIN**.

Table C3-2 shows the **AxDOMAIN[1:0]** signal encoding.

Table C3-2 Shareability domain encoding

AxDOMAIN[1:0]	Domain
`0b00`	Non-shareable
`0b01`	Inner Shareable
`0b10`	Outer Shareable
`0b11`	System

Restrictions apply to the shareability domain for transactions with different memory types:

- a *Device* transaction, indicated by **AxCACHE[1]** equal to zero, must only use domain level System
- a *Cacheable* transaction, indicated by **AxCACHE[3:2]** not equal to zero, must not use domain level System.

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Table C3-3 shows all **AxCACHE** and **AxDOMAIN** combinations. See *AXI4 changes to memory attribute signaling* on page A4-60 for details of the **AxCACHE** encodings.

Table C3-3 AxCACHE and AxDOMAIN signal combinations

AxCACHE		AxDOMAIN		Legal	Caches accessed[a]
Value	**Attribute**	**Value**	**Attribute**		
0b000x	Device	0b00	Non-shareable	No	-
		0b01	Inner Shareable	No	-
		0b10	Outer Shareable	No	-
		0b11	System	Yes	No caches.
0b001x	Non-cacheable	0b00	Non-shareable	Permitted	No caches.
		0b01	Inner Shareable	Permitted	Inner Shareable peer caches.
		0b10	Outer Shareable	Permitted	Outer Shareable peer caches.
		0b11	System	Yes	No caches.
0b011x 0b101x 0b111x	WriteThrough WriteBack	0b00	Non-shareable	Yes	Downstream caches.
		0b01	Inner Shareable	Yes	Inner Shareable peer caches and downstream caches.
		0b10	Outer Shareable	Yes	Outer Shareable peer caches and downstream caches.
		0b11	System	No	-

a. Shows which caches must be accessed to complete the transaction.

Note

- Table C3-3 does not include any access made to a local cache within the initiating master.
- The three combinations of **AxCACHE** and **AxDOMAIN** that are indicated as *Permitted* are legal within the protocol, but not expected. These combinations can be used when a memory location can be cached at a domain level that requires snooping, but the transaction is deliberately not cached downstream, for example, in a system level cache.
- When Table C3-3 shows that the caches accessed are Outer Shareable peer caches, this includes all caches that are Inner Shareable peer caches.
- A memory location that is indicated as being in the System domain cannot be held in any cache.

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C3.1.2 Read and write barrier transactions

Each address channel has its own barrier transaction signal. Table C3-4 shows the signal for each address channel.

Table C3-4 Barrier transaction signals

Transaction channel	Signal	Source	Description
Read address channel	**ARBAR[1:0]**	Master	Indicates a read barrier transaction.
Write address channel	**AWBAR[1:0]**	Master	Indicates a write barrier transaction.

See *Barrier transaction signaling* on page C8-249.

In this specification, **AxBAR** indicates **ARBAR** or **AWBAR**. Table C3-5 shows the **AxBAR[1:0]** signal encoding.

Table C3-5 Barrier transaction signal encoding

AxBAR[1:0]	Barrier type
`0b00`	Normal access, respecting barriers
`0b01`	Memory barrier
`0b10`	Normal access, ignoring barriers
`0b11`	Synchronization barrier

C3.1.3 Read and write shareable transaction types

Each address channel has its own transaction type signal. Table C3-6 shows the signal for each address channel.

Table C3-6 Shareable transaction type signals

Transaction channel	Signal	Source	Description
Read address channel	**ARSNOOP[3:0]**	Master	Indicates the transaction type of a read transaction.
Write address channel	**AWSNOOP[2:0]**	Master	Indicates the transaction type of a write transaction.

Transactions on the read and write address channel are categorized into the following groups:

Non-snooping These transactions must not snoop other masters.

Coherent These transactions are to memory locations that can be held in the cache of other masters and require snooping.

Memory update These transactions update main memory and must not snoop other masters.

Cache maintenance These transactions are to memory locations that can be held in the cache of other masters and require snooping. These transactions might also require to be passed to downstream caches.

Barrier These transactions establish the ordering between other transactions. See Chapter C8 *Barrier Transactions*.

DVM These transactions pass operations between components that participate in a distributed virtual memory scheme. See Chapter C12 *Distributed Virtual Memory Transactions*.

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Table C3-7 shows the permitted combinations of **ARBAR[0]**, **ARDOMAIN[1:0]**, and **ARSNOOP[3:0]** for each group of read transactions.

All unused **ARSNOOP[3:0]** encodings are Reserved.

Table C3-7 Permitted read address control signal combinations

Transaction group	ARBAR[0]	ARDOMAIN	ARSNOOP	Transaction type
Non-snooping	0b0	0b00 0b11	0b0000	ReadNoSnoop
Coherent	0b0	0b01 0b10	0b0000	ReadOnce
			0b0001	ReadShared
			0b0010	ReadClean
			0b0011	ReadNotSharedDirty
			0b0111	ReadUnique
			0b1011	CleanUnique
			0b1100	MakeUnique
Cache maintenance	0b0	0b00 0b01 0b10	0b1000	CleanShared
			0b1001	CleanInvalid
			0b1101	MakeInvalid
Barrier	0b1	0b00 0b01 0b10 0b11	0b0000	Barrier
DVM	0b0	0b01 0b10	0b1110	DVM Complete
			0b1111	DVM Message

Note

A component without a cache only needs to indicate the shareability of a read transaction using **ARDOMAIN** and can tie **ARSNOOP** to zero. As Table C3-7 shows, if the transaction is non-shareable it is treated as a ReadNoSnoop transaction and if the transaction is shareable it is treated as a ReadOnce transaction.

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Table C3-8 shows the permitted combinations of **AWBAR[0]**, **AWDOMAIN[1:0]**, and **AWSNOOP[2:0]** for each group of write transactions.

All unused **AWSNOOP[2:0]** encodings are Reserved.

Table C3-8 Permitted write address control signal combinations

Transaction group	AWBAR[0]	AWDOMAIN	AWSNOOP	Transaction type
Non-snooping	0b0	0b00 0b11	0b000	WriteNoSnoop
Coherent	0b0	0b01 0b10	0b000	WriteUnique
			0b001	WriteLineUnique
Memory update	0b0	0b00 0b01 0b10	0b010	WriteClean
			0b011	WriteBack
		0b01 0b10	0b100	Evict
Barrier	0b1	0b00 0b01 0b10 0b11	0b000	Barrier

Note

A component without a cache only needs to indicate the shareability of a write transaction using **AWDOMAIN** and can tie **AWSNOOP** to zero. As Table C3-8 shows, if the transaction is non-shareable it is treated as a WriteNoSnoop transaction and if the transaction is shareable it is treated as a WriteUnique transaction.

C3.1.4 Cache line size restrictions

The cache line size that each ACE master can support is determined at design time.

Restrictions apply to the minimum and maximum cache line sizes that can be supported.

The minimum cache line size is the larger of:

- 16 bytes
- the width of the data bus.

The maximum cache line size is the smaller of:

- 2048 bytes
- the product of the maximum burst length of 16 and the width of the data bus in bytes.

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Table C3-9 shows the minimum and maximum cache line sizes for each supported AXI4 data bus width.

Table C3-9 Minimum and maximum supported cache line sizes

Data bus width, bits	Minimum cache line size, bytes	Maximum cache line size, bytes
32	16	64
64	16	128
128	16	256
256	32	512
512	64	1024
1024	128	2048

C3.1.5 Transaction constraints

The following sections define the constraints for:

- *Cache line size transactions*
- *ReadOnce and WriteUnique transactions* on page C3-164
- *WriteBack and WriteClean transactions* on page C3-165
- *Barrier transactions* on page C3-166.

Cache line size transactions

The following transactions must be of cache line size:

- ReadClean
- ReadNotSharedDirty
- ReadShared
- ReadUnique
- CleanUnique
- MakeUnique
- CleanShared
- CleanInvalid
- MakeInvalid
- WriteLineUnique
- Evict.

Table C3-10 shows the constraints that apply to cache line size transactions.

Table C3-10 Cache line size transaction constraints

Attribute	Condition	Constraint
AxLEN	-	The burst length must be 1, 2, 4, 8 or 16 transfers. See *Burst length* on page A3-44.
AxSIZE	-	The number of bytes in a transfer must be equal to the data bus width. See *Burst size* on page A3-45.
AxBURST	INCR	The address must be aligned to the cache line size, which is equal to (**AxLEN** x **AxSIZE**), the total burst length in bytes. See *Burst type* on page A3-45.
	WRAP	The address must be aligned to **AxSIZE**, which is equal to the data bus width.
	FIXED	Not supported.

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Table C3-10 Cache line size transaction constraints (continued)

Attribute	Condition	Constraint
AxDOMAIN	All transactions except CleanShared, CleanInvalid, or MakeInvalid.	The domain must be Inner Shareable or Outer Shareable.
	CleanShared, CleanInvalid, or MakeInvalid transactions.	The domain must be Non-shareable, Inner Shareable or Outer Shareable.
AxBAR	-	Must be a normal access.
AxCACHE	-	Must be Modifiable.
AxLOCK	-	Must be: `0b0` If the transaction is ReadNotSharedDirty, ReadUnique, MakeUnique, CleanShared, CleanInvalid, MakeInvalid, WriteLineUnique or Evict. `0b0` or `0b1` If the transaction is ReadClean, ReadShared, or CleanUnique.
AxPROT	-	No constraint, can take any value.
AxQOS	-	No constraint, can take any value.

WriteLineUnique transactions are required to have every write data strobe asserted, that is, sparse write data strobes are not permitted.

The following transactions must use **AxLEN** to indicate the correct cache line size, even though these transactions do not transfer data:

- CleanUnique
- MakeUnique
- CleanShared
- CleanInvalid
- MakeInvalid
- Evict.

ReadOnce and WriteUnique transactions

The ReadOnce and WriteUnique transactions are not constrained to cache line size. This permits legacy components to operate in a coherent environment by adding an appropriate domain to Modifiable transactions.

Table C3-11 shows the constraints that apply to ReadOnce and WriteUnique transactions.

Table C3-11 ReadOnce and WriteUnique transaction constraints

Attribute	Constraint
AxDOMAIN	Must be Inner Shareable or Outer Shareable.
AxBURST	Must be INCR or WRAP.
AxCACHE	Must be Modifiable.
AxLOCK	Must be normal access.
AxPROT	No constraint, can take any value.
AxQOS	No constraint, can take any value.

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WriteUnique transactions are not required to have every write data strobe asserted, that is, sparse write data strobes are permitted.

Note

The FIXED burst type is not supported for ReadOnce and WriteUnique transactions. Any conversion from AXI to ACE-Lite must provide a translation for a FIXED burst.

WriteBack and WriteClean transactions

The WriteBack and WriteClean transactions are not constrained to cache line size. A partial update of a cache line is permitted. However, WriteBack and WriteClean transactions are constrained to updates within a single cache line.

Table C3-12 shows the constraints that apply to WriteBack and WriteClean transactions.

Table C3-12 WriteBack and WriteClean transaction constraints

Attribute	Condition	Constraint
AWLEN	-	Normal restrictions apply. See *Address structure* on page A3-44.
AWSIZE	-	Normal restrictions apply. See *Address structure* on page A3-44.
AWBURST	WRAP	The address must be aligned to **AxSIZE**, which is equal to the data bus width. The burst length must be 2, 4, 8 or 16. **AWSIZE** x **AWLEN** must not exceed the cache line size in bytes.
	INCR	The burst length must be 16 or less. The transaction must not cross a cache line boundary. The location of the last byte in the burst is determined by (**AWSIZE** x **AWLEN**) added to the **AWSIZE** aligned start address. The location of this last byte must fall within the same cache line as the first byte in the burst.
	FIXED	Not supported.
AWDOMAIN	-	Must not be domain type System.
AWBAR	-	Must be normal access.
AWCACHE	-	Must be Modifiable.
AWLOCK	-	Must be normal access.
AWPROT	-	No constraint, can take any value.
AWQOS	-	No constraint, can take any value.

The WriteBack and WriteClean transactions are permitted to use sparse write data strobes.

Components that support a snoop filter must correctly indicate the shareability domain for all WriteBack, WriteClean, and Evict transactions. This enables a snoop filter to track the allocation of Inner Shareable and Outer Shareable transactions.

A snoop filter must not track the allocation of non-shareable transactions because notification of the eviction of the associated cache line is not required.

Components that do not support a snoop filter can use any of the following domains for WriteBack or WriteClean transactions:

- Non-shareable
- Inner Shareable
- Outer Shareable.

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Barrier transactions

For a barrier transaction, as indicated by **AxBAR[0]** equal to 1, constraints apply to the read address and write address signals. Table C3-13 shows the constraints that apply:

Table C3-13 Barrier transaction constraints

Attribute	Constraint
AxADDR	Must be all zeros.
AxBURST	Must be burst type INCR.
AxLEN	Must be all zeros.
AxSIZE	Must be equal to the data bus width.
AxCACHE	Must be Normal, Non-cacheable.
AxPROT	No constraint, can take any value.
AxLOCK	Must be normal access.
AxSNOOP	Must be all zeros.

Note

A barrier transaction can have any shareability domain, and the choice of domain is used to determine the precise behavior of the barrier with respect to other masters in the system. See Chapter C8 *Barrier Transactions*.

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